

UNIVERSIDAD CENTRAL DEL ECUADOR



Criptografía y Seguridad de la información

Laboratorio Hacking Ético

Grupo 4

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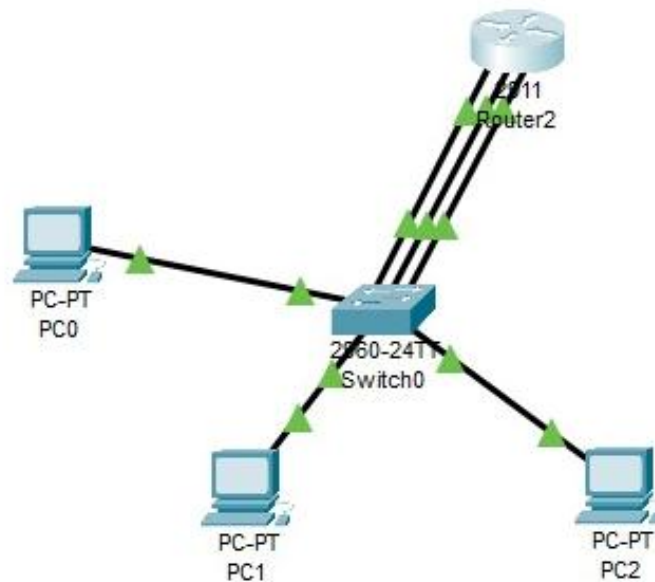
Olivares Cristian

Paca Belén

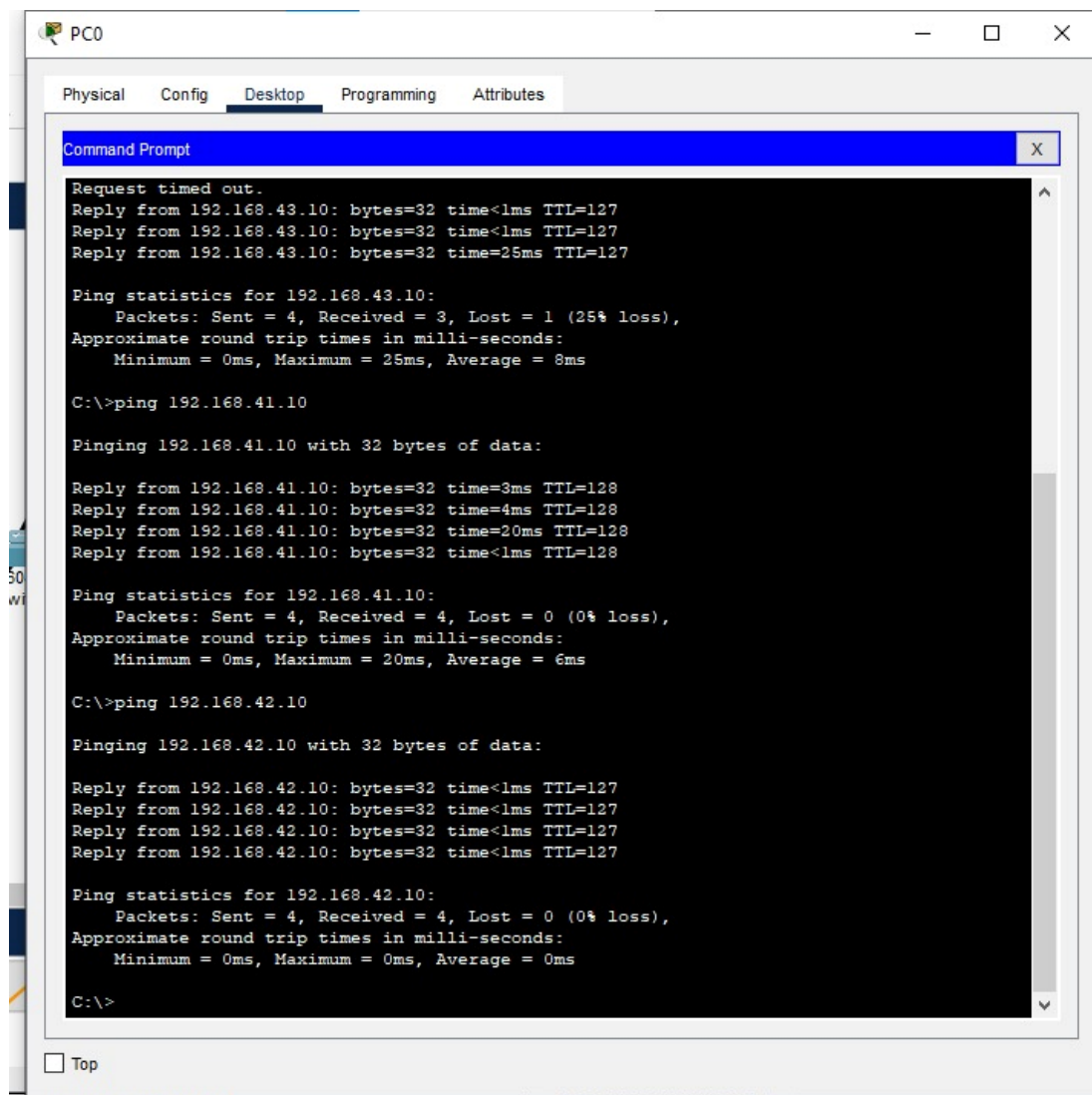
Introducción

El hacking ético es una práctica fundamental en el ámbito de la ciberseguridad que consiste en identificar y evaluar vulnerabilidades dentro de una red o sistema informático, con el propósito de fortalecer sus defensas. Esta actividad, realizada de manera controlada y con autorización, permite prevenir ataques maliciosos mediante la simulación de amenazas reales. En la presente práctica, se desarrolló una simulación de red que integra tres segmentos diferenciados (red1, red2 y red3), interconectados mediante dispositivos de red como rúter y switch Cisco, siguiendo una topología lógica con direcciones IP. Esta red permitió realizar pruebas controladas de herramientas de explotación disponibles en Kali Linux, enfocándonos en el escaneo de puertos con nmap y un ataque de denegación de servicio (DoS), con el fin de observar el comportamiento de los dispositivos frente a amenazas comunes.

Esquema de Red



Ping entre VLAN'S



The screenshot shows a PC0 Desktop window with a Command Prompt open. The Command Prompt displays the results of three ping commands. The first command is a ping to 192.168.43.10, which shows a 25% loss. The second command is a ping to 192.168.41.10, which shows 0% loss. The third command is a ping to 192.168.42.10, which also shows 0% loss.

```
Request timed out.  
Reply from 192.168.43.10: bytes=32 time<1ms TTL=127  
Reply from 192.168.43.10: bytes=32 time<1ms TTL=127  
Reply from 192.168.43.10: bytes=32 time=25ms TTL=127  
  
Ping statistics for 192.168.43.10:  
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 25ms, Average = 8ms  
  
C:\>ping 192.168.41.10  
  
Pinging 192.168.41.10 with 32 bytes of data:  
  
Reply from 192.168.41.10: bytes=32 time=3ms TTL=128  
Reply from 192.168.41.10: bytes=32 time=4ms TTL=128  
Reply from 192.168.41.10: bytes=32 time=20ms TTL=128  
Reply from 192.168.41.10: bytes=32 time<1ms TTL=128  
  
Ping statistics for 192.168.41.10:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 20ms, Average = 6ms  
  
C:\>ping 192.168.42.10  
  
Pinging 192.168.42.10 with 32 bytes of data:  
  
Reply from 192.168.42.10: bytes=32 time<1ms TTL=127  
Reply from 192.168.42.10: bytes=32 time<1ms TTL=127  
Reply from 192.168.42.10: bytes=32 time<1ms TTL=127  
Reply from 192.168.42.10: bytes=32 time<1ms TTL=127  
  
Ping statistics for 192.168.42.10:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\>
```

Direcciones y Puertos VLAN'S

- VLAN 1 (fa0/1-4)
 - Gateway: 192.168.41.1
 - Máscara de subred: 255.255.255.0
- VLAN 2 (fa0/5-8)
 - Gateway: 192.168.42.1
 - Máscara de subred: 255.255.255.0
- VLAN 3 (fa0/9-12)
 - Gateway: 192.168.43.1
 - Máscara de subred: 255.255.255.0

Configuración del SWITCH

Switch> enable

Switch# configure t

Switch(config)#interface range fa0/1-4

Switch(config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 1

Switch(config-if-range)#exit

Switch(config)#interface range fa0/5-8

Switch(config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 2

Switch(config-if-range)#exit

Switch(config)#interface range fa0/9-12

Switch(config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 3

Switch(config-if-range)#exit

Exclusión de las direcciones Gateway del servidor dhcp:

Switch(config)#ip dhcp excluded-address 192.168.41.1

Switch(config)#ip dhcp excluded-address 192.168.42.1

Switch(config)#ip dhcp excluded-address 192.168.43.1

Configuración de servidor dhcp para las vlan:

Switch(config)#ip dhcp pool vPool1

Switch(dhcp-config)#network 192.168.41.0 255.255.255.0

Switch(dhcp-config)#default-router 192.168.41.1

Switch(dhcp-config)#exit

```
Switch(config)#ip dhcp pool vPool12
Switch(dhcp-config)#network 192.168.42.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.42.1
Switch(dhcp-config)#exit
```

```
Switch(config)#ip dhcp pool vPool13
Switch(dhcp-config)#network 192.168.43.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.43.1
Switch(dhcp-config)#exit
```

Configuración puerta de enlace para las vlan:

```
Switch(config)#interface vlan 1
Switch(config-if)#ip add 192.168.41.1 255.255.255.0
Switch(config-if)#exit
```

```
Switch(config)#interface vlan 2
Switch(config-if)#ip add 192.168.42.1 255.255.255.0
Switch(config-if)#exit
```

```
Switch(config)#interface vlan 3
Switch(config-if)#ip add 192.168.43.1 255.255.255.0
Switch(config-if)#exit
```

Configuración del puerto FastEthernet para la comunicación con el router (trunk para tráfico de varias VLANs a través de un enlace):

```
Switch(config)#interface FastEthernet 0/1
```

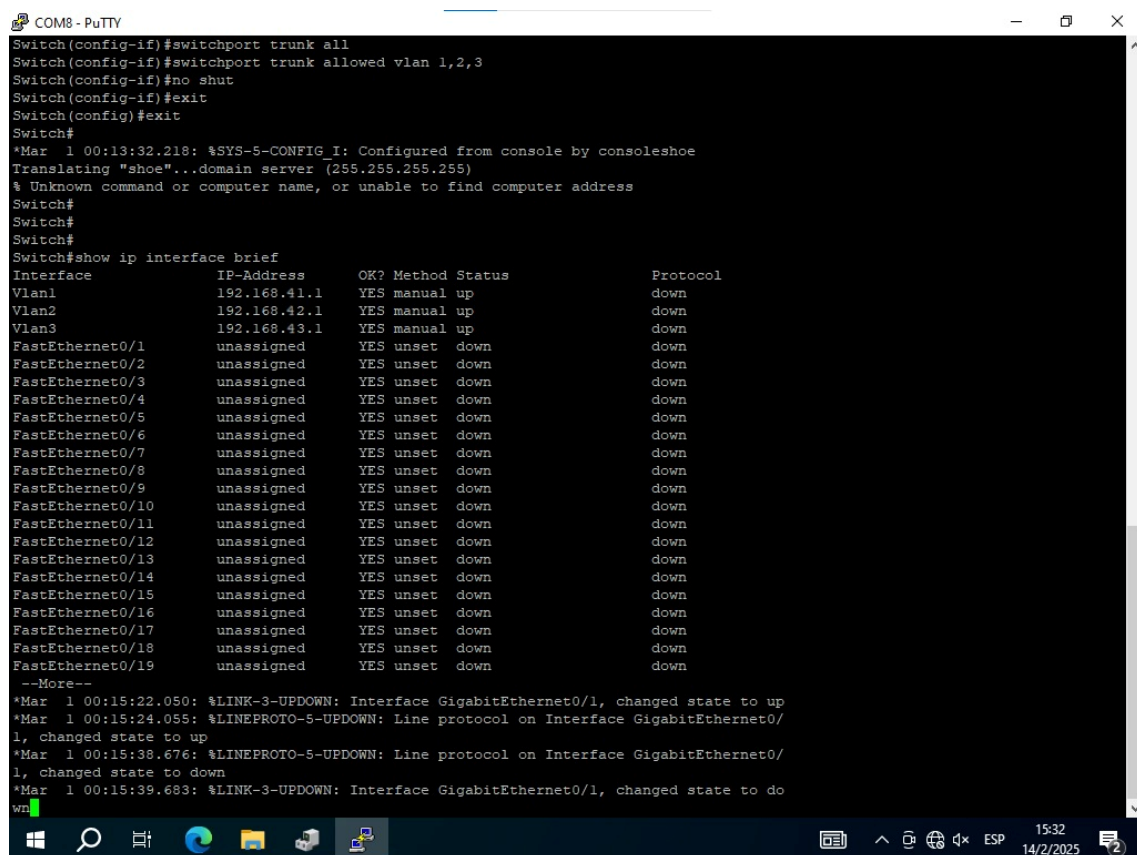
```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#switchport trunk allowed vlan 1,2,3
```

```
Switch(config-if)#no shut
```

```
Switch(config-if)#exit
```

Switch#show ip interface brief



```
COM8 - PuTTY
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 1,2,3
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#exit
Switch#
*Mar 1 00:13:32.218: %SYS-5-CONFIG_I: Configured from console by consoleshoe
Translating "shoe"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
Switch#
Switch#
Switch#
Switch#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Vlan1          192.168.41.1    YES manual up          down
Vlan2          192.168.42.1    YES manual up          down
Vlan3          192.168.43.1    YES manual up          down
FastEthernet0/1 unassigned      YES unset  down        down
FastEthernet0/2 unassigned      YES unset  down        down
FastEthernet0/3 unassigned      YES unset  down        down
FastEthernet0/4 unassigned      YES unset  down        down
FastEthernet0/5 unassigned      YES unset  down        down
FastEthernet0/6 unassigned      YES unset  down        down
FastEthernet0/7 unassigned      YES unset  down        down
FastEthernet0/8 unassigned      YES unset  down        down
FastEthernet0/9 unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  down        down
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
--More--
*Mar 1 00:15:22.050: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Mar 1 00:15:24.055: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
*Mar 1 00:15:38.676: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
*Mar 1 00:15:39.683: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to do
wn
```

CONFIGURACIÓN ROUTER

Configurar conexión (puerto) con el switch

```
Router> enable  
Router# configure t  
Router(config)#interface GigabitEthernet 0/0  
Router(config-if)#no ip address  
Router(config-if)#no shut  
Router(config-if)#exit
```

Configuración Subinterfaces

```
Router(config)#interface GigabitEthernet 0/0.1  
Router(config-subif)#encapsulation dot1Q 1  
Router(config-subif)#ip address 192.168.11.1 255.255.255.0  
Router(config-subif)#exit
```

```
Router(config)#interface GigabitEthernet 0/0.2  
Router(config-subif)#encapsulation dot1Q 2  
Router(config-subif)#ip address 192.168.12.1 255.255.255.0  
Router(config-subif)#exit
```

```
Router(config)#interface GigabitEthernet 0/0.3  
Router(config-subif)#encapsulation dot1Q 3  
Router(config-subif)#ip address 192.168.13.1 255.255.255.0  
Router(config-subif)#exit
```

Router#show ip interface brief

```
IOS Command Line Interface
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#show ip in
Router(config)#show ip interface brief
Router(config)#
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       192.168.41.1    YES manual up                    down
GigabitEthernet0/1       192.168.42.1    YES manual up                    down
GigabitEthernet0/2       192.168.43.1    YES manual up                    down
Vlan1                    unassigned      YES unset  administratively down  down

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router con0 is now available
```

CONFIGURACION ACCESS POINT

Después de reiniciar el AP se debe asignar la IP en el equipo que se vaya a configurar.

Propiedades de Habilitar el protocolo de Internet versión 4 (TCP/I... X

General

Puede hacer que la configuración IP se asigne automáticamente si la red admite esta funcionalidad. De lo contrario, deberá consultar con el administrador de red cuál es la configuración IP apropiada.

☐ Obtener una dirección IP automáticamente

☒ Usar la siguiente dirección IP:

Dirección IP: 192 . 168 . 1 . 10

Máscara de subred: 255 . 255 . 255 . 0

Puerta de enlace predeterminada: . . .

☐ Obtener la dirección del servidor DNS automáticamente

☒ Usar las siguientes direcciones de servidor DNS:

Servidor DNS preferido: . . .

Servidor DNS alternativo: . . .

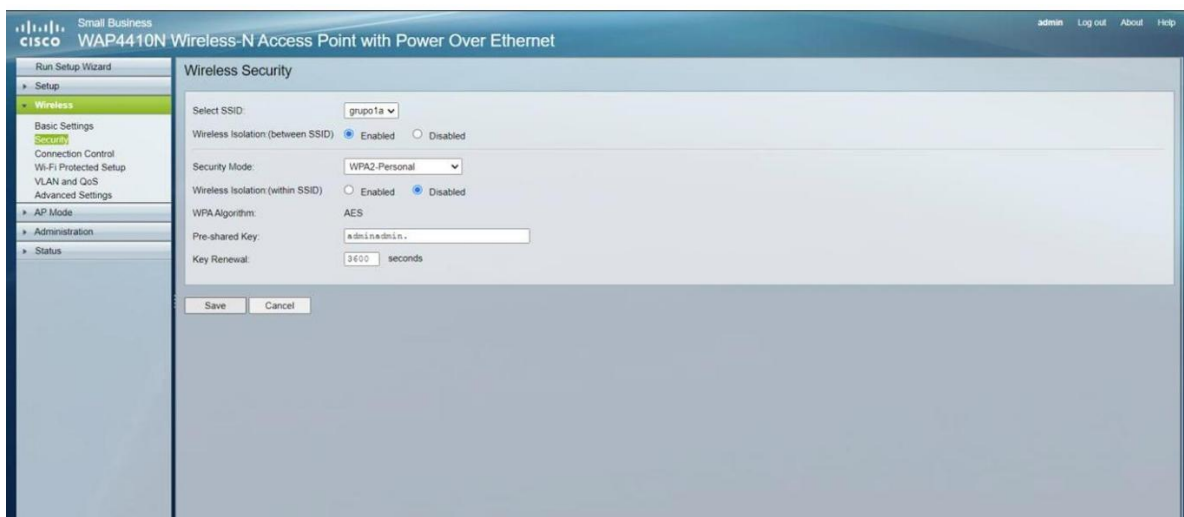
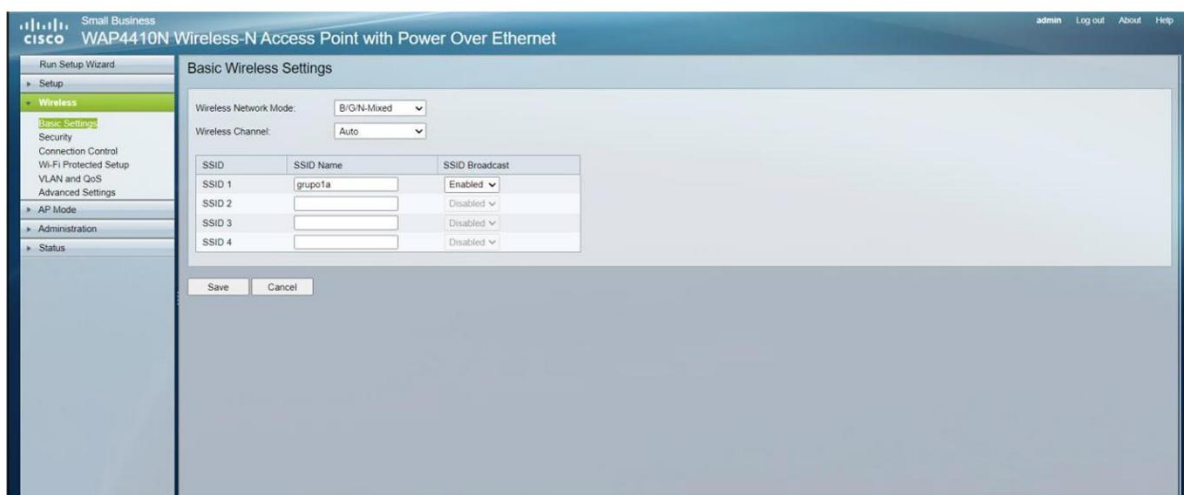
☐ Validar configuración al salir

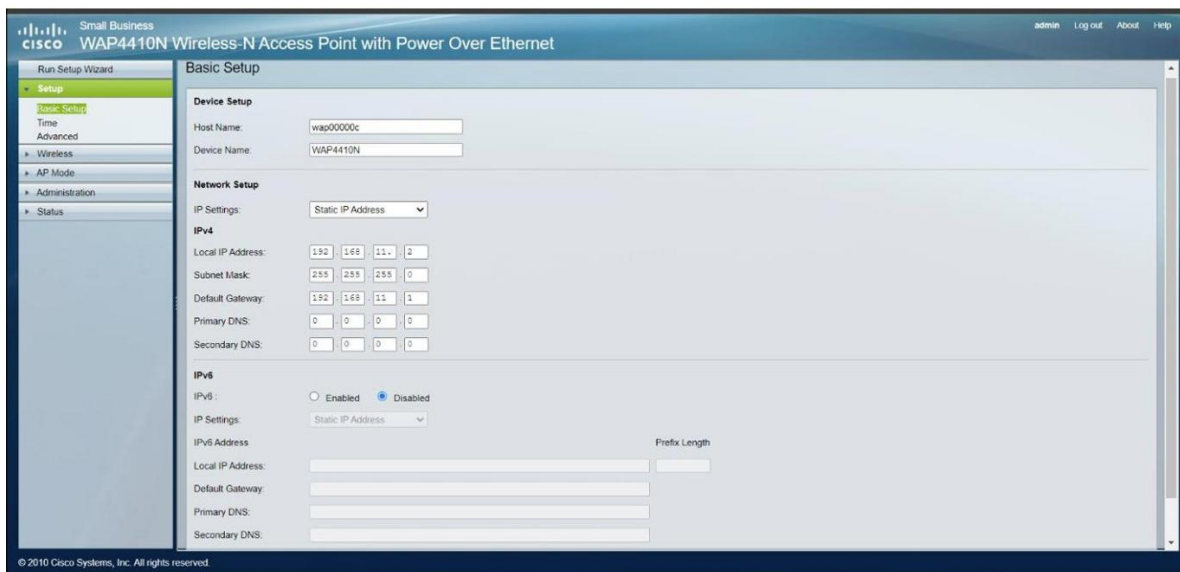
Opciones avanzadas...

Aceptar Cancelar

ingresamos al access point (AP) desde el equipo que acabamos de configurar mediante la IP 192.168.1.245

y configuramos la red WiFi:





ESCANEO DE PUERTOS (ABIERTO, CERRADO, FILTRADO)

Primero se verifico si hace ping con la otra maquina

```
kali@kali: ~
File Actions Edit View Help

valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
    inet 192.168.41.3/24 brd 192.168.41.255 scope global dynamic noprefixrout
e eth0
        valid_lft 86101sec preferred_lft 86101sec
        inet6 fe80::ab6e:ffdc:4eea:36a0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)~[~]
$ ping 192.168.41.1
PING 192.168.41.1 (192.168.41.1) 56(84) bytes of data.
64 bytes from 192.168.41.1: icmp_seq=2 ttl=255 time=12.4 ms
64 bytes from 192.168.41.1: icmp_seq=3 ttl=255 time=77.5 ms
64 bytes from 192.168.41.1: icmp_seq=4 ttl=255 time=99.8 ms
64 bytes from 192.168.41.1: icmp_seq=5 ttl=255 time=8.94 ms
64 bytes from 192.168.41.1: icmp_seq=6 ttl=255 time=5.83 ms
64 bytes from 192.168.41.1: icmp_seq=7 ttl=255 time=9.36 ms
64 bytes from 192.168.41.1: icmp_seq=8 ttl=255 time=6.12 ms
64 bytes from 192.168.41.1: icmp_seq=9 ttl=255 time=4.73 ms
64 bytes from 192.168.41.1: icmp_seq=10 ttl=255 time=4.92 ms
64 bytes from 192.168.41.1: icmp_seq=11 ttl=255 time=4.78 ms
64 bytes from 192.168.41.1: icmp_seq=12 ttl=255 time=15.6 ms
64 bytes from 192.168.41.1: icmp_seq=13 ttl=255 time=5.23 ms
```

```
kali@kali: ~  
File Actions Edit View Help  
64 bytes from 192.168.41.1: icmp_seq=14 ttl=255 time=5.77 ms  
64 bytes from 192.168.41.1: icmp_seq=15 ttl=255 time=7.43 ms  
64 bytes from 192.168.41.1: icmp_seq=16 ttl=255 time=12.1 ms  
64 bytes from 192.168.41.1: icmp_seq=17 ttl=255 time=6.40 ms  
64 bytes from 192.168.41.1: icmp_seq=18 ttl=255 time=6.87 ms  
64 bytes from 192.168.41.1: icmp_seq=19 ttl=255 time=6.60 ms  
^C  
— 192.168.41.1 ping statistics —  
19 packets transmitted, 18 received, 5.26316% packet loss, time 18056ms  
rtt min/avg/max/mdev = 4.727/16.684/99.840/25.879 ms  
  
(kali@kali)-[~]  
$ nmap 192.168.41.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-08 18:55 EDT  
Nmap scan report for 192.168.41.1  
Host is up (0.055s latency).  
Not shown: 996 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
23/tcp    open  telnet  
80/tcp    open  http  
443/tcp   open  https  
MAC Address: 68:BC:0C:40:E3:40 (Cisco Systems)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.02 seconds  
  
(kali@kali)-[~]  
$
```

Después se usó NMAP se escaneo los puertos

```
kali-linux-2025.2-virtualbox-amd64 [Corriendo] - Oracle VM VirtualBox  
Archivo Máquina Ver Entrada Dispositivos Ayuda  
1 2 3 4  
File Actions Edit View Help  
root@kali: /home/kali  
$ sudo ifconfig  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST> MULTICAST UP-LINK-UP-etu 1500 qlisc fq_codel state UP g  
rout default qlen 1000  
link/ether 08:00:27:0d:01:5d brd ff:ff:ff:ff:ff:ff  
inet 192.168.41.1/24 brd 192.168.41.255 scope global dynamic noprefixrou  
t  
valid_lft 84403sec preferred_lft 84403sec  
inet 192.168.1.1/24 scope global eth0  
valid_lft forever preferred_lft forever  
inet6 fe80::a002:70d:1:5d%4 scope link noprefixroute  
valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
$ nmap 192.168.1.10  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-08 19:43 EDT  
Nmap scan report for 192.168.1.10  
Host is up (0.0025s latency).  
Not shown: 994 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
2906/tcp  open  snmp  
5137/tcp  open  wsddapi  
5432/tcp  open  postgresql  
MAC Address: 08:9E:01:71:5F:DE (Quanta Computer)  
  
Nmap done: 1 IP address (1 host up) scanned in 15.11 seconds  
  
(kali@kali)-[~]  
$ sudo apt  
root@kali: /home/kali  
$ apt install hping3  
hping3 is already the newest version (3.a2.ds2-11-kali1).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
  
(root@kali)-[/home/kali]  
$ hping3 -S -flood -p 1000 192.168.1.10  
WARNING: 192.168.1.10 (to 192.168.1.10): 5 set, 40 headers + 0 data bytes  
hping in flood mode, no results will be shown  
^C  
— 192.168.1.10 hping statistic —  
590090 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms  
  
(root@kali)-[/home/kali]
```

ATAQUE DE DENEGACIÓN DE SERVICIO

Usando HPING se realizó el ataque de denegación de servicio

```
root@kali: /home/kali
File Actions Edit View Help

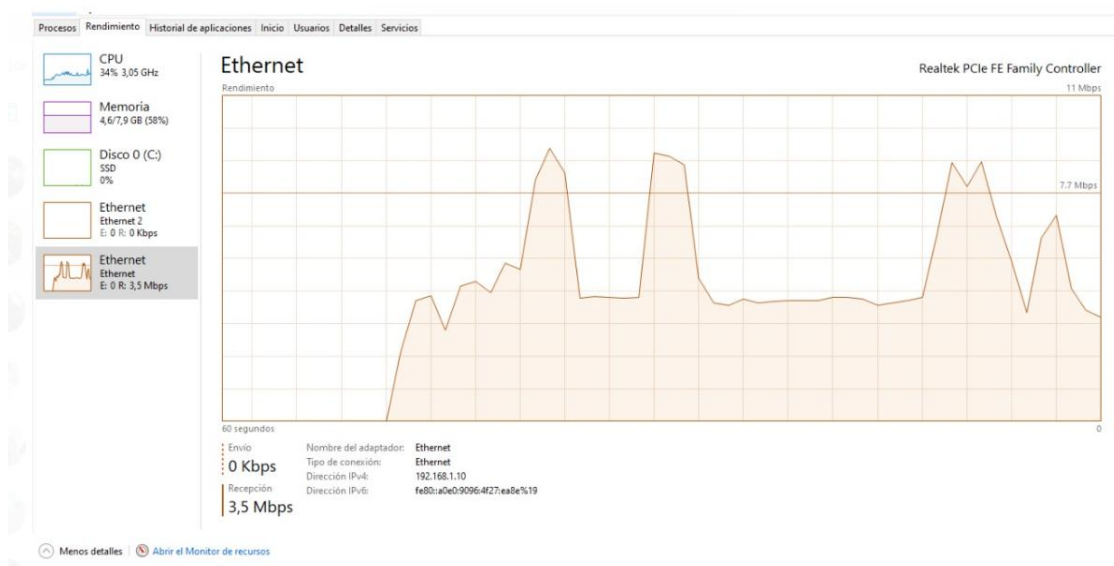
(kali@kali)-[~]
$ hping3 -S --flood -p 80 192.168.41.4
[open_sockraw] socket(): Operation not permitted
[main] can't open raw socket

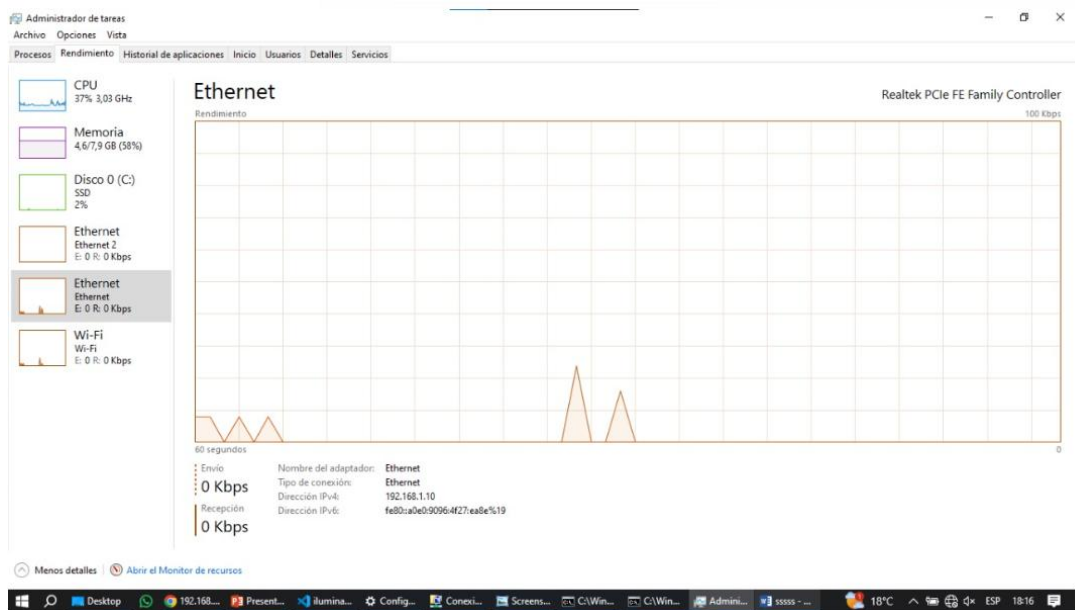
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# hping3 -S --flood -p 192.168.41.1
hping3: missing host argument
Try 'hping3 --help' for more information.

(root@kali)-[/home/kali]
# hping3 -S --flood -p 1400 192.168.41.1
HPING 192.168.41.1 (eth0 192.168.41.1): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.41.1 hping statistic —
573677 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(root@kali)-[/home/kali]
# hping3 -S --flood -p 1400 192.168.1.10
HPING 192.168.1.10 (eth0 192.168.1.10): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Desde el equipo

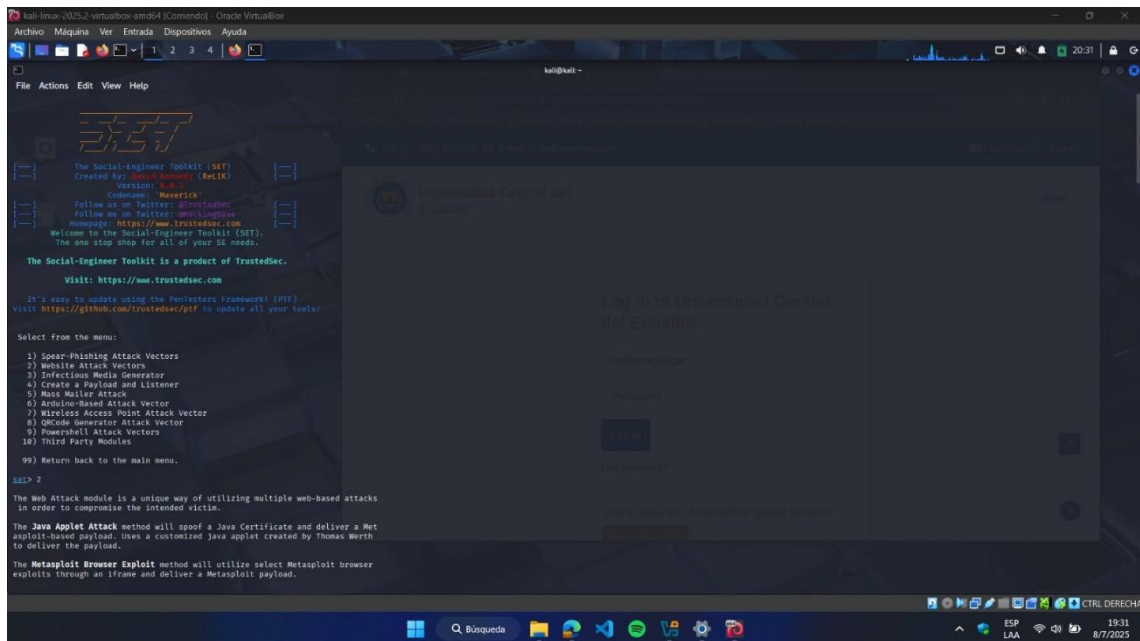




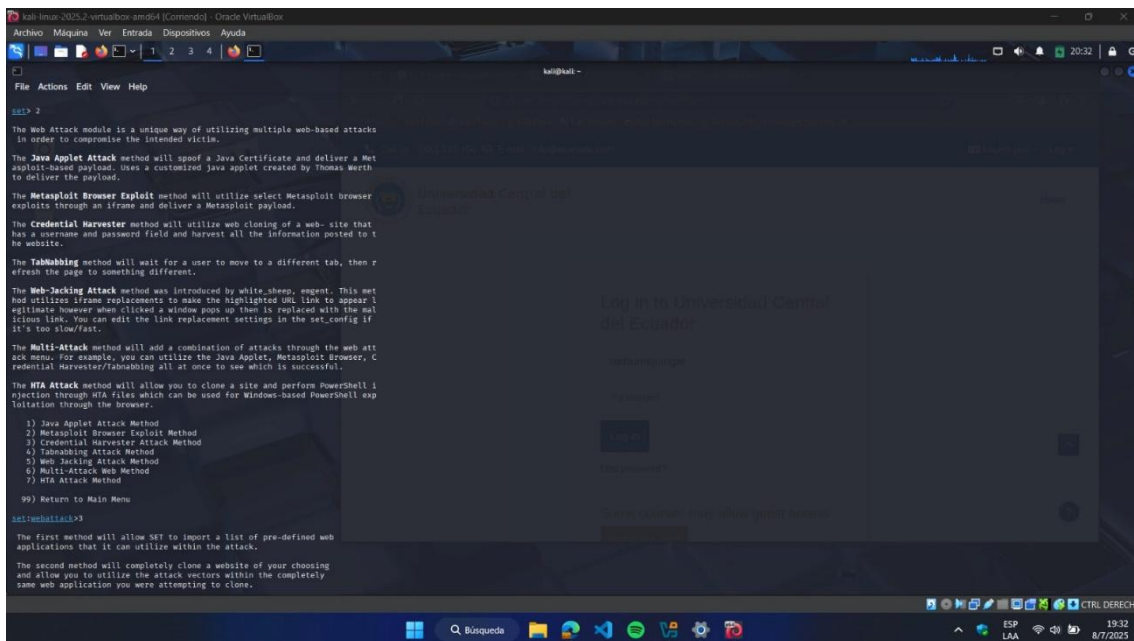
UNA ACTIVIDAD ABIERTA (LA QUE EL GRUPO CONSIDERE)

Para este caso hemos elegido hacer uso de las herramientas de ingeniería social en Kali en especial Toolkit

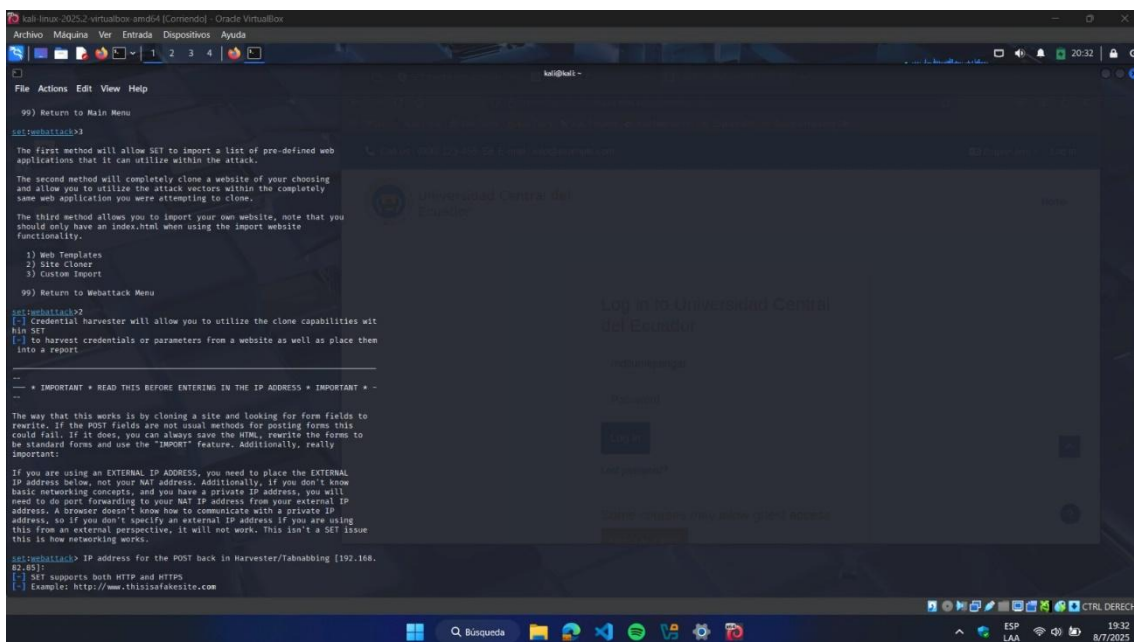
\$ sudo setoolkit



Elegimos la opción 2) Website Attack



Elegimos la opción 3) Credential Harvester Attack Method



Elegimos la opción 2) Site cloner

Como podemos observar una vez ingresamos a la IP ahí se encuentra clonada la página y el usuario al ingresar sus credenciales estas se mostrarán en la consola.

